

Effect of Network Topology

NodeBasedModels.jl Vignette 02

Simon Frost

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Introduction

Population-level pairwise models characterise a network solely through summary statistics — the mean degree $\langle k \rangle$, the second moment $\langle k^2 \rangle$, and possibly the clustering coefficient ϕ . Node-based models, by contrast, can use any graph instance, so we can directly investigate how the precise topology shapes the epidemic.

In this vignette we compare three canonical network families — regular, Erdős–Rényi, and scale-free (Barabási–Albert) — and examine special cases (trees, cycles, complete graphs) where theoretical guarantees exist.

Setup

```

using NodeBasedModels
using Graphs
using Plots
using OrdinaryDiffEqDefault
using Random
using Statistics

```

Regular vs Erdős–Rényi vs Scale-free

We create three graphs on $N = 200$ nodes, each targeting a mean degree of approximately 6.

```

Random.seed!(123)

g_reg = random_regular_graph(200, 6)
g_er = erdos_renyi(200, 6 / 199)
g_ba = barabasi_albert(200, 3) # each new node adds 3 edges → mean degree ≈ 6

net_reg = GraphNetwork(g_reg)
net_er = GraphNetwork(g_er)
net_ba = GraphNetwork(g_ba)

println("Regular — nodes: ", nv(g_reg), ", edges: ", ne(g_reg),
        ", ⟨k⟩ = ", round(mean_degree(net_reg); digits=2))
println("Erdős–Rényi — nodes: ", nv(g_er), ", edges: ", ne(g_er),
        ", ⟨k⟩ = ", round(mean_degree(net_er); digits=2))
println("Barabási–Albert — nodes: ", nv(g_ba), ", edges: ", ne(g_ba),
        ", ⟨k⟩ = ", round(mean_degree(net_ba); digits=2))

```

```

Regular — nodes: 200, edges: 600, ⟨k⟩ = 6.0
Erdős–Rényi — nodes: 200, edges: 586, ⟨k⟩ = 5.86
Barabási–Albert — nodes: 200, edges: 591, ⟨k⟩ = 5.91

```

Degree distributions

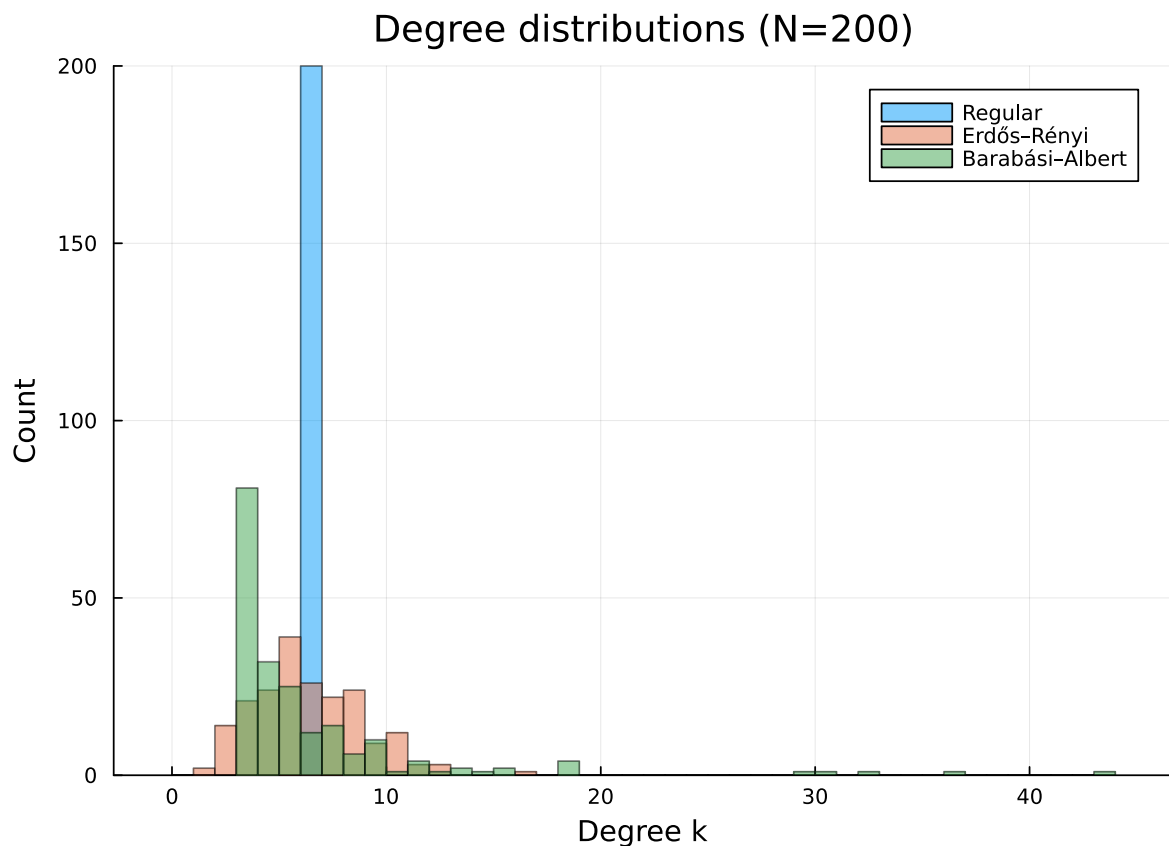
The three families have very different degree distributions despite similar mean degree. The regular graph is a delta function at $k = 6$, the Erdős–Rényi graph is approximately Poisson, and the Barabási–Albert graph has a heavy (power-law) tail.

```

deg_reg = degree(g_reg)
deg_er = degree(g_er)
deg_ba = degree(g_ba)

p = histogram(deg_reg, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Regular",
              xlabel = "Degree k", ylabel = "Count",
              title = "Degree distributions (N=200)")
histogram!(p, deg_er, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Erdős-Rényi")
histogram!(p, deg_ba, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Barabási-Albert")
p

```



Pair-based dynamics on each topology

We run the pair-based (order-2) model on each graph with the same epidemiological parameters.

```

 $\tau$  = 0.125
 $\gamma$  = 0.25

pb_reg = generate_pair_based(sir_model(), net_reg;
    infection_rate =  $\tau$ , recovery_rate =  $\gamma$ ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)
pb_er = generate_pair_based(sir_model(), net_er;
    infection_rate =  $\tau$ , recovery_rate =  $\gamma$ ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)
pb_ba = generate_pair_based(sir_model(), net_ba;
    infection_rate =  $\tau$ , recovery_rate =  $\gamma$ ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)

I_reg = aggregate(pb_reg, :I)
I_er = aggregate(pb_er, :I)
I_ba = aggregate(pb_ba, :I)
t = range(0.0, 40.0, length = length(I_reg))

```

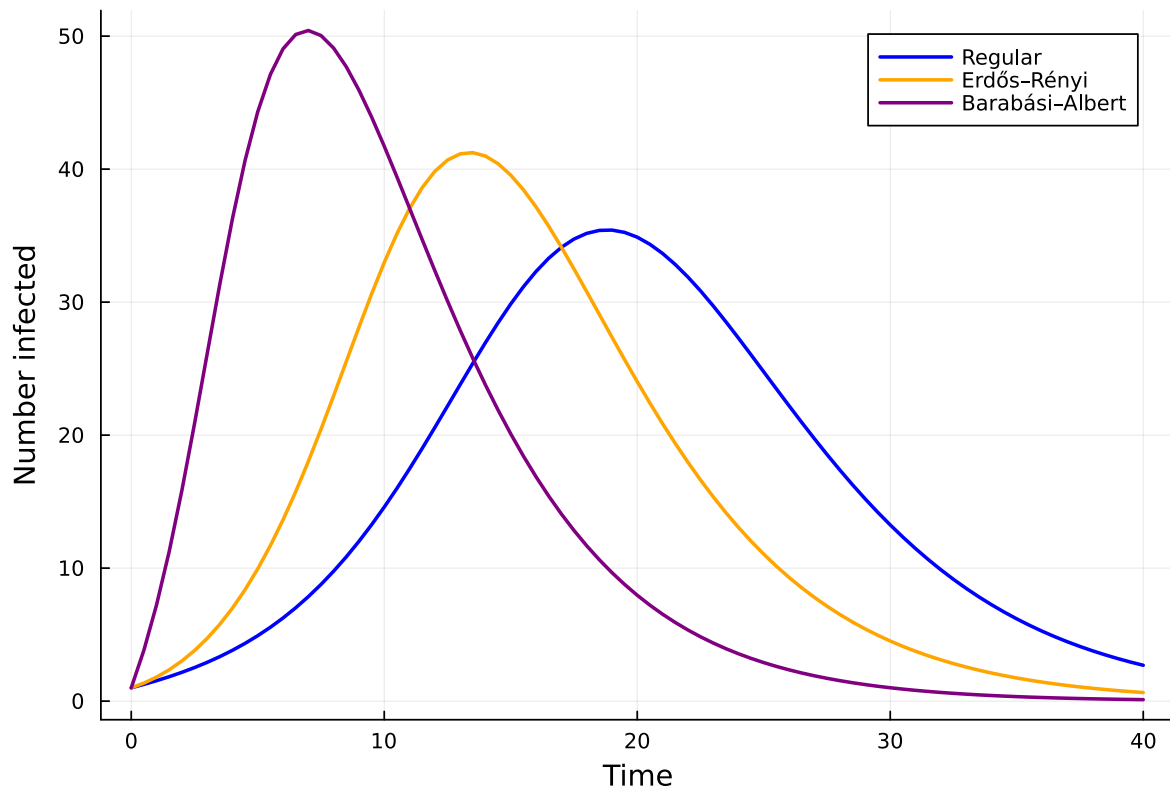
0.0:0.5:40.0

```

p = plot(t, I_reg, label = "Regular", lw = 2, color = :blue,
    xlabel = "Time", ylabel = "Number infected",
    title = "Pair-based SIR: effect of topology (N=200, R0=2 anchor, k=6)")
plot!(p, t, I_er, label = "Erdős-Rényi", lw = 2, color = :orange)
plot!(p, range(0.0, 40.0, length = length(I_ba)), I_ba,
    label = "Barabási-Albert", lw = 2, color = :purple)
p

```

Pair-based SIR: effect of topology (N=200, R₀=2 anchor, k=6)



The Barabási-Albert (scale-free) graph typically shows the fastest early growth, driven by its high-degree hub nodes. These hubs are infected early and rapidly spread infection to many neighbours. The regular graph, with its uniform degree, produces the most “classical” epidemic curve.

Epidemic threshold

For population-level pairwise models on a homogeneous (regular) network with degree n and the Bernoulli closure, the epidemic threshold is

$$\tau_c = \frac{\gamma}{n-2}.$$

This depends only on the degree, not on the specific graph realisation:

```
tau_c = epidemic_threshold(regular_network(6), BernoulliClosure(), gamma)
println("Epidemic threshold tau_c = ", round(tau_c; digits=4))
println("Our tau = $tau -> R_0 proxy above threshold: tau > tau_c = ", tau > tau_c)
```

Epidemic threshold $\tau_c = 0.0625$
Our $\tau = 0.125 \rightarrow R_0$ proxy above threshold: $\tau > \tau_c = \text{true}$

The basic reproduction number at the population level:

```
R0 = basic_reproduction_number(regular_network(6), BernoulliClosure(),  $\tau$ ,  $\gamma$ )
println("R0 (Bernoulli, n=6) = ", round(R0; digits=3))
```

R_0 (Bernoulli, n=6) = 2.0

Tree vs cyclic graph

On a tree graph the pair-based (Kirkwood) closure is exact because every pair of neighbours of a node i are conditionally independent given i — there are no short cycles.

We compare a tree with a cycle graph of the same number of nodes.

```
Random.seed!(99)
n_small = 30
g_tree = prufer_decode(rand(1:n_small, n_small - 2))
g_cycle = cycle_graph(n_small)

net_tree = GraphNetwork(g_tree)
net_cycle = GraphNetwork(g_cycle)

println("Tree - edges: ", ne(g_tree), ", mean degree: ",
        round(mean_degree(net_tree); digits=2))
println("Cycle - edges: ", ne(g_cycle), ", mean degree: ",
        round(mean_degree(net_cycle); digits=2))
```

Tree - edges: 29, mean degree: 1.93
Cycle - edges: 30, mean degree: 2.0

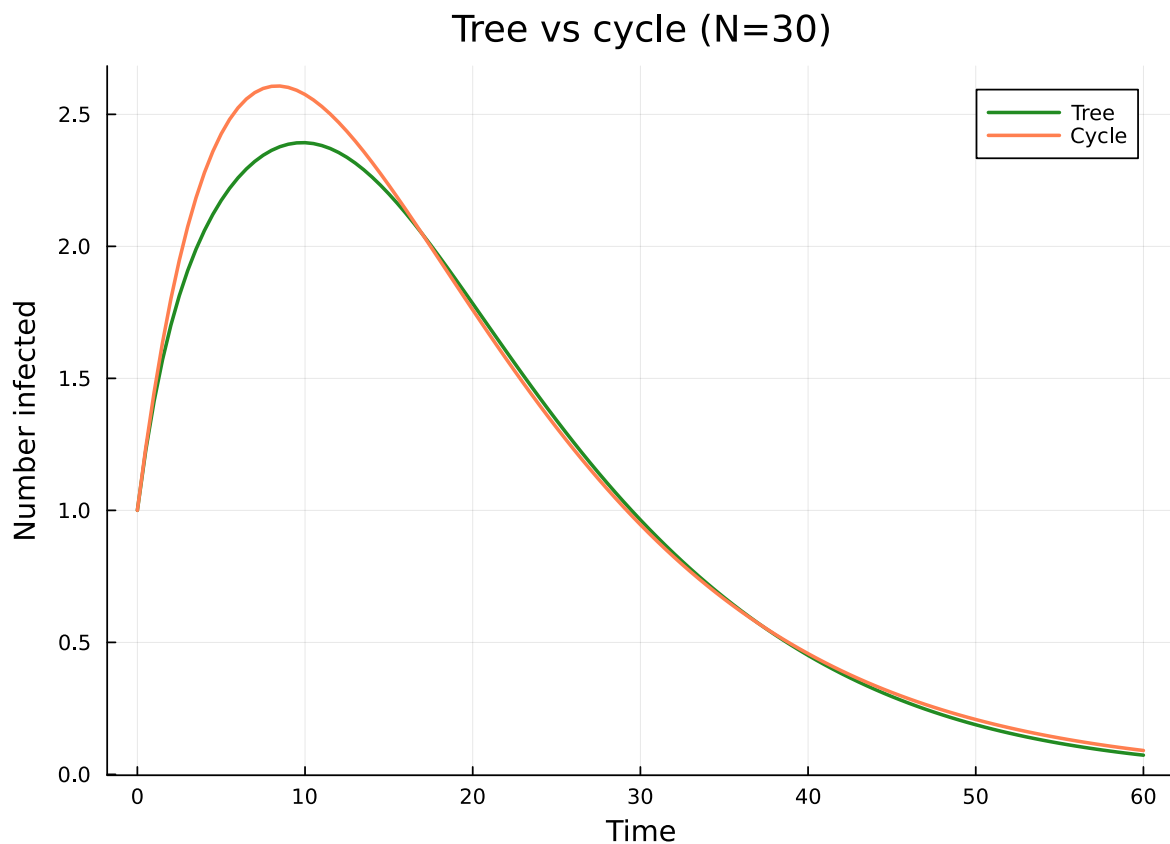
```
pb_tree = generate_pair_based(sir_model(), net_tree;
    infection_rate = 0.3, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 60.0), saveat = 0.5)
pb_cycle = generate_pair_based(sir_model(), net_cycle;
    infection_rate = 0.3, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 60.0), saveat = 0.5)
```

```

I_tree = aggregate(pb_tree, :I)
I_cycle = aggregate(pb_cycle, :I)
t_small = range(0.0, 60.0, length = length(I_tree))

p = plot(t_small, I_tree, label = "Tree", lw = 2, color = :forestgreen,
         xlabel = "Time", ylabel = "Number infected",
         title = "Tree vs cycle (N=$n_small)")
plot!(p, range(0.0, 60.0, length = length(I_cycle)), I_cycle,
      label = "Cycle", lw = 2, color = :coral)
p

```



The tree, despite having similar mean degree, supports faster epidemic spread because the branching structure allows infection to reach many nodes simultaneously, unlike the cycle where infection can only travel in one direction.

Complete graph: recovering mass-action

On a complete graph (K_N), every node is connected to every other node. In this limit the individual-based (NIMFA) model should closely approximate the classical mass-action SIR ODE, since the graph is maximally homogeneous and degree variance is zero.

```
g_comp = complete_graph(50)
net_comp = GraphNetwork(g_comp)

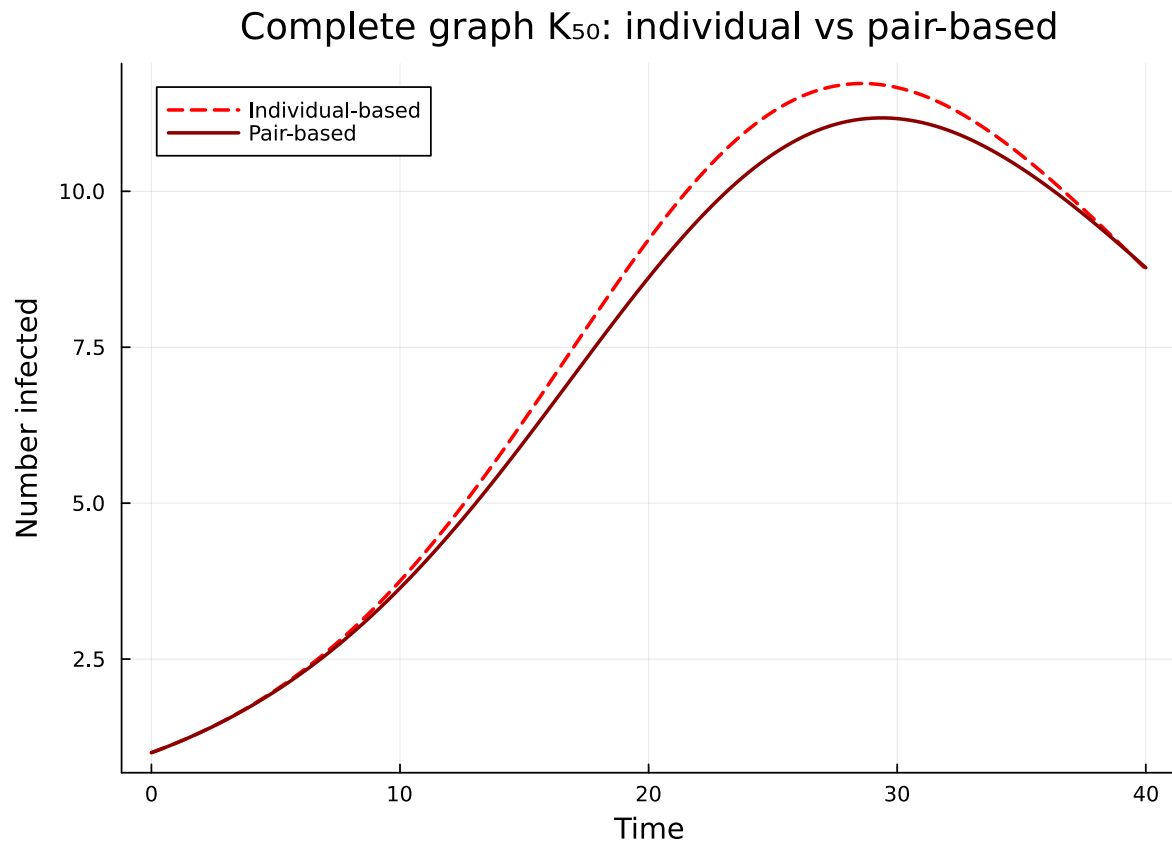
ib_comp = generate_individual_based(sir_model(), net_comp;
    infection_rate = 0.005, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.25)

pb_comp = generate_pair_based(sir_model(), net_comp;
    infection_rate = 0.005, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.25)

S_ib = aggregate(ib_comp, :S)
I_ib = aggregate(ib_comp, :I)
S_pb = aggregate(pb_comp, :S)
I_pb = aggregate(pb_comp, :I)

t_comp = range(0.0, 40.0, length = length(S_ib))

p = plot(t_comp, I_ib, label = "Individual-based", lw = 2, ls = :dash, color = :red,
    xlabel = "Time", ylabel = "Number infected",
    title = "Complete graph  $K_{50}$ : individual vs pair-based")
plot!(p, range(0.0, 40.0, length = length(I_pb)), I_pb,
    label = "Pair-based", lw = 2, color = :darkred)
p
```

On the complete graph, the individual-based and pair-based models are very close because the independence assumption becomes nearly exact: the neighbours of any node i form an almost-complete subgraph, so conditioning on i 's state provides little extra information about its neighbours.

Summary

Topology	Key property	Effect on epidemic
Regular	Uniform degree	Predictable, moderate speed
Erdős–Rényi	Poisson degree	Moderate heterogeneity
Barabási–Albert	Power-law degree (hubs)	Fastest early growth, lower threshold
Tree	No cycles	Pair-based model is exact
Cycle	Minimal connectivity	Slowest spread
Complete	Maximum connectivity	Individual-based $\approx$ mass-action

Network structure controls three key quantities:

1. Epidemic speed — Hubs accelerate early spread (scale-free networks).
2. Final epidemic size — Degree heterogeneity generally increases the final size.
3. Model accuracy — The pair-based closure is exact on trees; the individual-based closure is best on complete (or near-complete) graphs. On graphs with many short cycles, neither may suffice and Gillespie simulation is needed.

NetworkOutbreaks SSA ribbon

We overlay the `NetworkOutbreaks.jl` Gillespie SSA on each topology to confirm that the pair-based prediction sits inside the stochastic ensemble. Because $N = 200$ is small here, each ribbon is averaged over multiple graph realisations from the same family. The seed fraction is matched to the deterministic single-node seed ($\varepsilon = 1/200 = 0.005$).

```
include("../_validation.jl")

N_v = 200
ε_v = 1 / N_v
prog_no = sir_model(τ = :β)
params_no = Dict{:β ⇒ τ, :γ ⇒ γ}

t_reg, μ_reg, σ_reg = gillespie_ribbon(prog_no, params_no,
    regular_graph_builder(N_v, 6);
    N = N_v, n_graphs = 6, nsims_per_graph = 20,
    tspan = (0.0, 40.0), seed_fraction = ε_v)
t_er, μ_er, σ_er = gillespie_ribbon(prog_no, params_no,
    poisson_graph_builder(N_v, 6.0);
    N = N_v, n_graphs = 6, nsims_per_graph = 20,
    tspan = (0.0, 40.0), seed_fraction = ε_v)
t_ba, μ_ba, σ_ba = gillespie_ribbon(prog_no, params_no,
    barabasi_albert_graph_builder(N_v, 3);
    N = N_v, n_graphs = 6, nsims_per_graph = 20,
    tspan = (0.0, 40.0), seed_fraction = ε_v)
```

```
([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 ... 35.5, 36.0, 36.5, 37.0, 37.5, 38.0,
```

```
p = plot(t, I_reg, label = "Regular (PB)", lw = 2, color = :blue,
    xlabel = "Time", ylabel = "Number infected",
    title = "Pair-based vs NetworkOutbreaks SSA across topologies")
plot!(p, t_reg, μ_reg[:I] .* N_v, ribbon = σ_reg[:I] .* N_v,
    label = "Regular (SSA)", color = :blue,
    fillalpha = 0.2, linealpha = 0.5, lw = 1)
```

```

plot!(p, t, I_er, label = "Erdős-Rényi (PB)", lw = 2, color = :orange)
plot!(p, t_er, μ_er[:I] .* N_v, ribbon = σ_er[:I] .* N_v,
      label = "Erdős-Rényi (SSA)", color = :orange,
      fillalpha = 0.2, linealpha = 0.5, lw = 1)
plot!(p, range(0.0, 40.0, length = length(I_ba)), I_ba,
      label = "Barabási-Albert (PB)", lw = 2, color = :purple)
plot!(p, t_ba, μ_ba[:I] .* N_v, ribbon = σ_ba[:I] .* N_v,
      label = "Barabási-Albert (SSA)", color = :purple,
      fillalpha = 0.2, linealpha = 0.5, lw = 1)
p

```

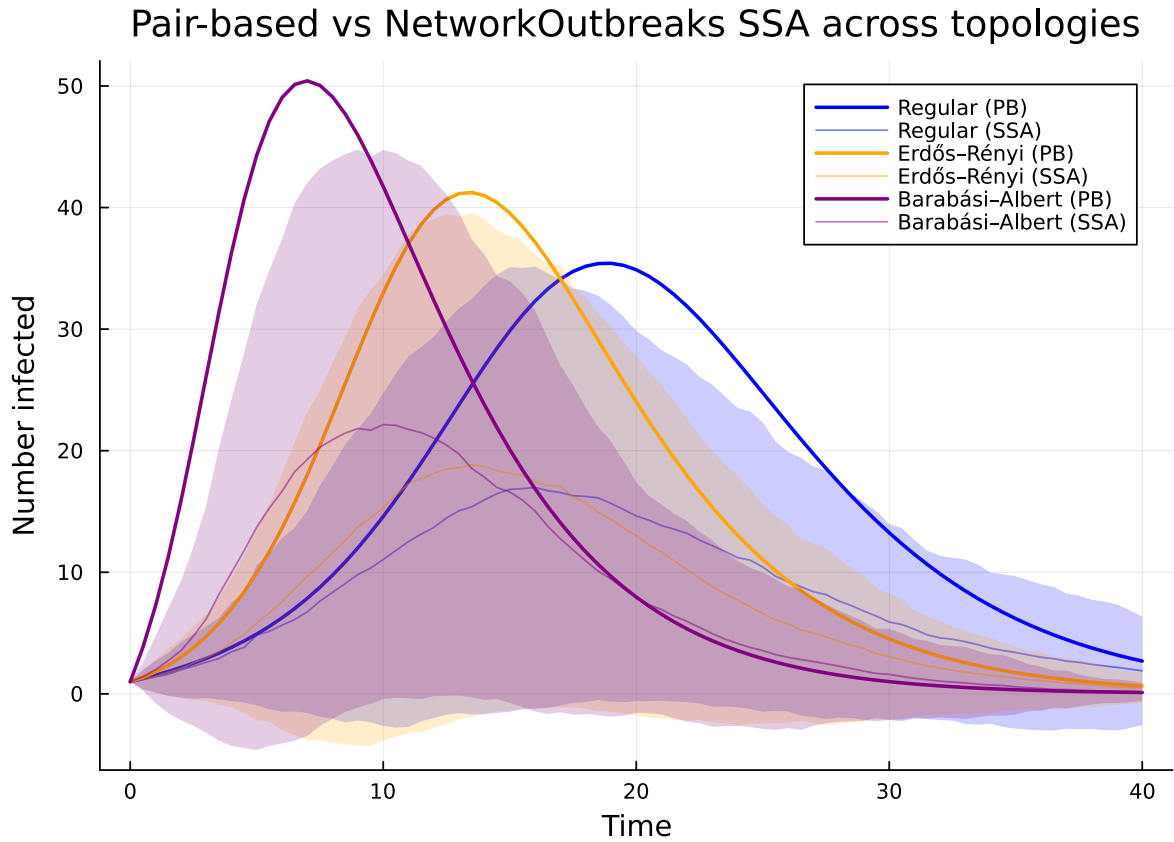


Figure 1: Pair-based deterministic predictions (lines) vs NetworkOutbreaks SSA mean $\pm 1\sigma$ ribbons (shaded). Each ribbon averages 6 graph realisations $\times$ 20 stochastic runs at $N=200$, $\varepsilon=1/200$.